Ahmed Zein Elabedin *Mobile Application Developer*

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Profile

Senior mobile engineer with 5 years of experience building production apps with Dart and Flutter, currently shipping for the UAE market. I work across the full lifecycle — architecture, implementation, CI/CD, and store release — and go deep where most Flutter work stops: three-isolate background execution, real-time WebSocket sync, iOS Live Activities and Dynamic Island, and custom platform channels into Kotlin and Swift. I built a real-time voice assistant on the Gemini Live API with bidirectional audio streaming, and authored my platform's engineering handbook covering architecture decision records, production incident history, and team onboarding. I publish and maintain open-source Flutter packages on pub.dev used by third-party developers, and run one of them in production in a commercial app. Earlier work includes a Firestore re-architecture that cut ~829M reads per year and made invoice creation ~99% faster on a system handling over 3.5M AED monthly. I review code, mentor junior developers, and have delivered remote and on-site work with teams across the UAE, UK, Sweden, and Egypt. My apps are live on the App Store and Google Play.

Professional Experience

2025/04 – present
Abu Dhabi, United Arab Emirates • Onsite

Senior Software Engineer
Kief Consultancy & Technology

- Lead mobile engineer across three production Flutter applications — Taggo, Taggo Captain, and The Events — covering architecture, implementation, release management, and mentoring.

Taggo — multi-service mobility platform (iOS & Android)

- Built the customer app for a four-service ecosystem: ride-hailing and cargo delivery live, with car rental and car recovery built and waiting on their backend.
- Implemented real-time driver tracking over Socket.IO, with a background service on Android that keeps following the trip while the app is in the background.
- Shipped iOS Live Activities and Dynamic Island support so active trip status stays visible without opening the app.
- Built **Taggo Buddy**, a real-time voice assistant on the Gemini Live API: bidirectional audio streaming over WebSocket (16 kHz capture, 24 kHz playback), push-to-talk, mixed text+binary frame demultiplexing, and reconnect with backoff and mid-session auth refresh. Users book rides by voice through a draft→confirm→commit flow gated by biometric authentication.
- Built the in-app wallet and payments on Stripe — saved cards, the payment sheet, Apple Pay and Google Pay, and 3D Secure for payments and top-ups.
- Delivered phone sign-in with backend OTP, plus Google and Apple sign-in via Firebase Auth, and full Arabic/English localisation with RTL.

Taggo Captain — driver companion app (iOS & Android)

- Built the real-time offer system: WebSocket dispatch with an accept/reject screen, and background alerts on both platforms — a full-screen alert that wakes the screen on Android, a push notification on iOS.
- Implemented continuous background location reporting with quality filtering, heartbeat and idle downgrade, ping buffering, and self-healing recovery.
- Shipped iOS Live Activities surfacing rider, ETA, and earnings.
- Built the navigation map: a heading-up driving camera with speed-based zoom and smooth car movement that holds its heading when the car stops.

- Automated driver onboarding with on-device document scanning (licence, passport, residency) using document_camera_frame — my own open-source package, running in production here.

The Events — e-commerce & gifting (iOS & Android)

- Modernised a legacy Flutter codebase: refactored for maintainability, introduced consistent state management, and unified UI/UX.
- Integrated payment gateways including Telr, Google Pay, and Apple Pay, with PDF generation and printing for order documents.
- Implemented forced app updates, secure storage, dependency injection, pagination, and HTML content rendering.
- **Cut app launch time from ~12 s to ~3 s on the same device**: lazy-loaded home sections and tabs, cached and de-duplicated provider requests, a native splash, shimmer skeletons and SVG pre-caching.

Across Taggo and Taggo Captain

- Designed a **three-isolate background architecture** (main, background service, FCM handler) with explicit cross-isolate contracts — conflating them is the root of most background-execution bugs.
- Hardened both apps for release: **freeRASP device integrity** that blocks rooted or jailbroken devices and wipes the local session, screenshot and screen-recording blocking on sensitive screens (sign-in, wallet, payments, cash-out), Keychain/Keystore token storage, disabled backups, Dart obfuscation, and release logs free of location data and tokens.
- Built **release guards on every build path** (a CLI check, a Gradle task, an Xcode archive script and a CI workflow) that block a release missing its security configuration or signed with the debug key, because a missing freeRASP config silently ships an unprotected app.
- Replaced **certificate pinning**, which Cloudflare's unannounced certificate rotation would have turned into an outage, with system-CA-only trust on Android and App Transport Security on iOS.
- Set up **dev and prod flavors** in both apps, with separate staging and production backends, app IDs and Firebase apps, installable side by side. One entry point reads the flavor at compile time and falls back to production rules. Caught a Retrofit code-generation trap that could have pointed staging builds at production.
- Built **payment reliability engineering**: 3D Secure, idempotency keys, settlement reconciliation, recovery from ambiguous failures, plus wallet, driver cash-out, and cash-deposit flows.
- Own **dual-store release engineering**: a version-baseline ledger, single-version releases to both stores, validated multi-locale release notes, rollback, and tester distribution — including Play Store compliance work for 16 KB page size and target API level upgrades.
- Authored the **platform engineering handbook** — architecture decision records, production incident history, and developer guide — as the team's onboarding reference.
- Conduct in-depth code reviews, **mentor junior developers**, and set architecture standards (Clean Architecture, BLoC/Cubit).
- Work directly with product owners to turn requirements into technical specifications.

2024/05 – 2025/04
Ras Al-Khaimah, United Arab Emirates • Onsite

Mid-Senior Software Engineer
BA3.co
BA3 Business Solution — enterprise accounting and business management across Android, iOS, Windows, macOS, and Web from a single Flutter codebase; 26 modules handling over 3.5M AED in monthly transactions.

- Re-architected invoice fetching with a linked-list structure, eliminating full-collection reads and saving ~760M Firestore reads/year; a hybrid material cache saved a further ~68M. Combined **~829M reads/year (≈$497)**, with invoice creation **~99% faster**.
- Built a custom desktop windowing layer — floating, draggable, priority-stacked windows and dialog stacks — replicating native behaviour Flutter desktop doesn't provide, with focus management for operators running several tasks at once. Later rewrote it and open-sourced it as draggable_floating_window.
- Delivered role-based access control with granular seller/admin permissions and real-time multi-platform sync via Firestore listeners.
- Implemented year-end migration: balance rollover, closing-to-opening stock carry-forward, unpaid cheque handling, and secure account closure.
- Localised into six languages (Arabic, English, Urdu, German, French, Chinese) with RTL and culture-aware formatting.
- Integrated WhatsApp API for instant invoice sharing, plus JSON/XML import-export and direct printing.
- Built NFC-based attendance check-in/out and Bluetooth thermal printer support for POS receipts.

Amber Drive — customer and provider apps for a car rental service (iOS & Android).

- Integrated Stripe Connected Accounts for automated provider payouts, with webhook handling that keeps payments reliable when the app is interrupted mid-transaction.
- Built Google Maps location-based vehicle search with navigation and live availability.
- Created document_camera_frame, a custom Flutter package for licence/passport/residency capture, removing a third-party SDK dependency and its cost.
- Implemented a multi-status reservation workflow (pending, approved, current, finished, cancelled) with real-time Firebase sync and push notifications.

2024/02 – 2024/05
Project-Based Contract
London, United Kingdom • Remote

Mid-Senior Mobile App Developer
Gainz.ai
AI-powered fitness app delivering personalised workout plans based on goals, fitness level, and preferences.

- Built the Notepad feature set, including search across note content, categories, and coaches with matched-text highlighting and scroll-to-match navigation.
- Implemented on-device AI operations (summarisation and translation) and optimised search and data retrieval for responsiveness.
- Published **Scroll Highlight Text** to pub.dev, extracted from this work.
- Designed the Note Categories and Note Details screens with a focus on UI/UX, smooth transitions, and quick response times.

"*Ahmed Zein is a skilled Flutter developer with a strong work ethic and problem-solving abilities. His expertise in Flutter development, coupled with effective communication and collaboration skills, makes him a valuable asset to any team.*" — Yousef Shams, Mobile Team Lead, GAINZ.AI

2022/01 – 2024/01
Cairo, Egypt • Onsite

Mobile Application Developer
Xtreme Digital
- Led mobile application development across the full lifecycle, collaborating with cross-functional teams to optimise performance and reduce delivery time.
- Successfully launched over 8 apps on the App Store and Google Play, including Maharaja Restaurants, IBM Pharma, Fitn Kitchen, and Ikoo Media.

2021/07 – 2021/12
Project-Based Contract
Stockholm, Sweden • Remote

Flutter Developer
LinnStäd
- Delivered the full mobile app lifecycle from concept to App Store launch.

- Maintained and enhanced the LinnStäd Cleaning Services app, now serving **over 10,000 active users**.

Projects

Taggo
A multi-service mobility platform for the UAE market, combining ride-hailing and cargo delivery, with car rental and car recovery built behind a launch flag until their backend is ready. Real-time driver tracking fed by the driver app's background GPS on both platforms, iOS Live Activities and Dynamic Island for live trip status, and an in-app wallet on Stripe. Includes Taggo Buddy, a voice assistant on the Gemini Live API that books rides through streamed conversation.
Live on both stores — Google Play: play.google.com/store/apps/details?id=ae.kief.taggo

Taggo Captain
The driver-side companion app for the Taggo platform. Drivers receive and manage ride offers in real time over WebSockets, navigate with a heading-up driving map, track daily, weekly and monthly earnings, and cash out to their bank account. Onboarding scans licences, passports, and residency cards using document_camera_frame, my own open-source package.
Live on both stores — Google Play: play.google.com/store/apps/details?id=ae.kief.taggodriver.taggo_driver

The Events
Events is the leading online flowers and gifting company in the United Arab Emirates and Arabian Gulf. Founded in 2022, it expanded within three years to nine countries, sourcing products daily from ethical vendors, artists, and celebrities to create an experience that fits every occasion. I modernised the Flutter codebase and delivered payments (Telr, Google Pay, Apple Pay), PDF and print, secure storage, forced updates, and cut app launch time from about 12 seconds to 3.
Live on both stores — Google Play: play.google.com/store/apps/details?id=com.logicalyinfotech.events

BA3 Business Solution
Cross-platform (Android, iOS, Windows, macOS, Web) business system with advanced billing, financial management (accounts, bonds, cheques), inventory tracking, and role-based access control. Custom floating windows replicate native desktop behaviour (now open source as draggable_floating_window); modular architecture with reusable widgets for scalability and efficient maintenance.

Amber Drive
A car rental marketplace for the UAE, built as one Flutter app with separate renter and owner flows. Renters sign in with UAE Pass, phone or biometrics, scan their driving licence with an on-device OCR camera frame (where document_camera_frame started), find cars on a custom-styled map with routes, and book with a slide-to-book gesture. Payments run on Stripe, with Stripe Connect payouts to car owners. Owners manage their fleet, approve bookings that update on the renter’s phone in real time, and file damage reports with photos.

Gainz AI: Chat, Train, Track
Next-gen fitness app featuring hyper-realistic AI athlete avatars and adaptive training plans, delivering real-time coaching, personalised workouts and nutrition, and comprehensive progress tracking.
Live on both stores — Google Play:play.google.com/store/apps/details?id=ai.gainz.app

Osta Home
Commercial real estate marketplace integrating residential, commercial, and service construction. Users negotiate directly with verified experts, view ratings and prices, and manage schedules; registered brokers receive official IDs.

Spark Gym
Spark Fitness & Healthcare Center offers personalised group training in supportive environments, with tailored activities closely monitored by coaches.

Al-Taher Jewelry
Two apps: a client app for discovering gold jewellery with real-time prices and search, and a manager app for store operations — jewellers, products, and daily gold prices.

Open Source Contributions

Draggable Floating Window
Desktop-style windows inside a Flutter app: draggable, resizable, priority-stacked windows with per-window dialog stacks, keyboard focus memory and shortcuts, and Flutter's own dialogs and menus working above them. Rewritten from the windowing layer I built for BA3, with no dependencies beyond Flutter, 160/160 pub points, and an integration test that drives the real macOS app.

Document Camera Frame
A customisable camera interface for capturing and cropping document images. Supports both CocoaPods and Swift Package Manager, native VNDocumentCameraViewController scanning on iOS, OCR and text extraction via Google ML Kit, five built-in UI modes, and export to JPG, PNG, PDF, or TIFF. **Running in production in Taggo Captain** for driver document onboarding, and maintained against issues filed by third-party adopters.

Scroll Highlight Text
Highlights and scrolls through text content in both Latin and Arabic, with customisable highlighting and scroll-to-match navigation.

Skills

Languages & Core
Dart · Flutter · Kotlin · Swift

Architecture & State
Clean Architecture · OOP & SOLID · design patterns · algorithms & data structures · flutter_bloc (Cubit + Bloc) · equatable · dartz functional error handling · get_it dependency injection · Provider · GetX

Background Execution
Three-isolate architecture (main / background service / FCM) · flutter_background_service · Android foreground services · iOS background location + APNs · Isolates, Compute & Streams · heartbeat with idle downgrade · GPS quality filtering · self-healing recovery

Flutter Desktop
In-app windowing (draggable_floating_window) · custom ModalRoutes & nested Navigators · focus scopes & focus restoration · priority-based z-order · Shortcuts & Actions · ThemeExtension theming

Security
freeRASP device integrity (root/jailbreak detection) · flutter_secure_storage · data-at-rest hardening · API key restriction · Firebase SHA fingerprints · secret management & build-time guards

Maps & Location
google_maps_flutter · markers, geolocation & routing · GeoJSON polygon rendering · geolocator, geocoding, permission_handler

UI & Localisation
Responsive & adaptive UI · flutter_screenutil · animations · easy_localization with enforced key parity · RTL support

Quality & Process
Unit, widget & integration testing · debugging & profiling tools · Architecture Decision Records · production incident postmortems · technical documentation & team onboarding · Git & GitHub · Jira, Trello, Slack, Asana · Figma · Postman

AI & Real-Time Audio
Gemini Live API integration via backend relay · bidirectional audio streaming over WebSocket (16 kHz / 24 kHz) · push-to-talk · mixed text+binary frame demultiplexing · on-device summarisation & translation

Networking & Realtime
Dio · Retrofit · json_serializable codegen · REST & GraphQL · Socket.IO / WebSockets — dual-socket architecture, reconnection policy, handshake auth

Native Platform Integration
Custom platform channels (Kotlin / Swift) · iOS Live Activities & Dynamic Island · Android notification channels & native screen-wake · deep linking (cold & warm launch paths)

Payments
Stripe (incl. Connected Accounts & webhooks) · Telr · PayPal · Razorpay · Google Pay & Apple Pay · 3D Secure / SCA · idempotency keys · settlement reconciliation · wallet & cash-out flows

Firebase & Backend
Authentication · Firestore · Realtime Database · Storage · FCM · Cloud Functions · Hosting · Node.js

Data & Storage
Hive · Sqflite · Shared Preferences · PDF generation & printing · JSON/XML import-export

Release Engineering
Flutter flavors · CocoaPods & Swift Package Manager · Gradle · CI/CD with GitHub Actions & Fastlane · App Store Connect & Play Console · version baseline ledger · dual-store single-version releases · multi-locale release notes · rollback · Firebase App Distribution · Play Store compliance (16 KB page size, target API levels)

Education

2017 – 2021 **Bachelor's degree - Faculty of Telecommunication Engineering and Computer Department**
Delta University For Science And Technology

Certificates

- Stanford University Algorithms Specialization
- University of California San Diego Object Oriented Programming in Java
- Udacity one million arab coders initiative
- Algorithms Analysis and Design from scratch
- Data Structures Decode
- Node.js From Scratch